YUANKAI TAO

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SUMMARY

Current master student in Operations Research program in Georgia Institute of Technology, Isye, with research experience in large-scale spatial data, machine learning, statistical modeling, and computational optimization. Research interests center on reliable, scalable methods at the intersection of machine learning, data science, and operations research.

RESEARCH INTERESTS

Machine Learning and Deep Learning; Statistical Learning; Large-Scale Data Science; Optimization and Simulation; Time-Series Modeling; Risk and Decision Analytics; Computational Methods

EDUCATION

Georgia Institute of Technology *Aug 2026 – Dec 2027*
M.S. in Operations Research, Stewart School of Industrial and Systems Engineering
Selected Coursework: Deterministic Optimization, Computational Statistics, Simulation

University of California, San Diego *Sep 2024 – Dec 2025*
M.S. in Quantitative Finance, Rady School of Management *GPA:* 3.84/4.0
Selected Coursework: Deep Learning, Computational Finance Methods, Data Science for Finance

Zhongnan University of Economics and Law *Sep 2020 – Jun 2024*
B.S. in Management, School of Accounting and Finance *GPA:* 3.63/4.0
Selected Coursework: Probability & Statistics, Linear Algebra, Calculus, Python Programming

RESEARCH EXPERIENCE

Research Assistant for Professor Franklin Qian, UNC-Chapel Hill *Mar 2025 – Jan 2026*
The Numbers Game: Signaling Effects of Pricing Formats in Housing Bargaining

• Engineered a reproducible Python, PostgreSQL, and PostGIS pipeline for a 76-million-observation spatial dataset with 40+ variables on Google Cloud Platform.

• Implemented parallelized spatial computations for comparable-housing variables across multiple distance ranges and property-specific filters, achieving approximately 50× computational speedup.

• Built and queried a 964.5M-record, 100GB+ database with Python and DuckDB on UNC's Longleaf high-performance computing cluster; extracted population statistics under time-window and property-level constraints.

SELECTED PROJECTS

Kaggle Jane Street Market Prediction Competition *Silver Medal*

• Built a memory-efficient Polars/PyTorch pipeline for 47M intraday observations across 1,669 trading days; engineered lagged, rolling, and temporal features and deployed online inference through Kaggle's streaming evaluation API.

• Implemented a parameter-efficient TabM regressor with a 32-member shared-parameter ensemble; one-hot encoded three categorical features and improved cross-validation performance by 0.004.

• Contributed to the baseline training and final ensemble of XGBoost, MLP, Ridge, and TabM, helping the team place 94th of 3,757 teams.

Text Analysis and BERT Fine-tuning on SEC 10-K Reports *MGTF 423 Final Project, UCSD*
Team Lead | Human-in-the-Loop Learning on 20.5M Text Rows

• Conducted named-entity recognition, frequency analysis, and exploratory text analysis on SEC filings.

• Designed a three-round human-in-the-loop fine-tuning workflow for scarce labels; trained a BERT model achieving 79% accuracy in quantifying uncertainty tone in corporate disclosures.

Trajectory Prediction Challenge, UCSD CSE 251B *Argoverse 2 Dataset*

• Applied coordinate rotation, flipping, and Gaussian-noise augmentation to trajectory data and developed an autoregressive OneStepLSTM with rolling features.

• Improved the test metric from 10.22 to 8.50 compared with a traditional decision-tree baseline.

UCSD / FMSbonds, Inc. 2025 Capstone Project *Aug 2025 – Dec 2025*

Team Lead | High-Performance Bond Pricing Engine Optimization

• Designed a caching system for high-frequency cash-flow queries, reducing per-bond solving time from 1.17 ms to 0.13 ms (9× speedup).

• Rewrote the yield-to-worst numerical solver with Numba JIT, reducing computation time from 1.57 ms to 7.5 μs per bond (210× speedup).

PROFESSIONAL EXPERIENCE

Guangdong Hedge Fund Investment Management Co., Ltd. *May 2026 – Jul 2026*

Quantitative Research Intern *Quantitative Strategy Department*

• Engineered an end-to-end commodity-futures ML pipeline with 16 trend, momentum, volatility, liquidity, volume, and open-interest features; trained LightGBM, XGBoost, CatBoost, and neural-network models across 1- to 20-day horizons.

• Used rolling out-of-sample validation and statistical, risk, and transaction-cost metrics; implemented a monthly rebalanced strategy with a 120-day retraining window that achieved a 1.89 Sharpe ratio, 16.79% annualized return, and 25.51% excess return after transaction costs.

• Built a dynamic VaR/CVaR framework with 17 model specifications combining volatility models and distributional assumptions to study long/short momentum risk.

VariFlight Technology *Jul 2025 – Sep 2025*

Data Analyst Intern *Data Development Department*

• Built Python web-scraping, OCR, and table-extraction workflows that transformed 600+ unstructured PDF reports and 500+ webpages into standardized datasets for downstream local LLM training.

• Created dashboards for aviation transportation and passenger-turnover indicators, supporting structured analysis of operational trends.

TECHNICAL SKILLS

Programming & Systems	Python, SQL, MATLAB, LaTeX, Git, Linux, GCP, HPC
Quantitative Methods	Statistical Learning, Time-Series Analysis, Optimization, Simulation
Libraries & Frameworks	Pandas, Polars, Numpy, Scipy, PyTorch, scikit-learn, CVXPy, Backtrader

HONORS & CERTIFICATIONS

WorldQuant Alphathon 2024, Gold Level (Rank 16, Top 3.1%) *2024*

Beta Gamma Sigma (Top 20% of graduating class) UCSD, *2024 – 2025*

Deep Learning Specialization (5-Course Certificate) Coursera, DeepLearning.AI

Algorithms and Data Structures Certificate (6-Course Certificate) Coursera, UC San Diego

Special Scholarship in Innovation and Entrepreneurship ZUEL, *2022 – 2023*